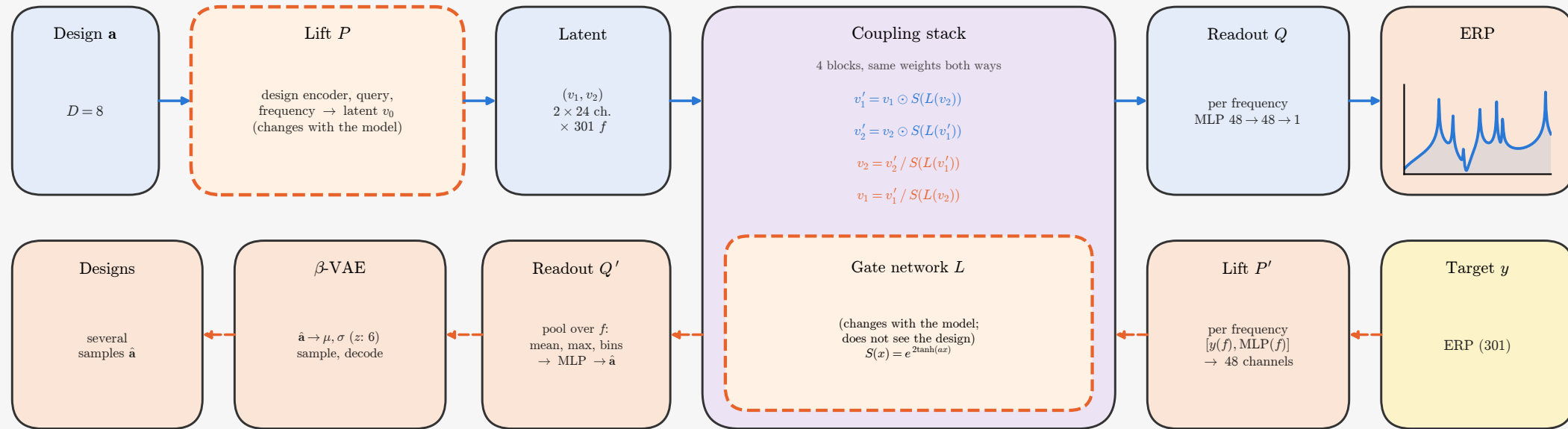


Invertible coupling-flow operators: common structure

one coupling stack run forwards (design $\rightarrow$ ERP) and backwards (ERP $\rightarrow$ designs); the lift P and the gate L change



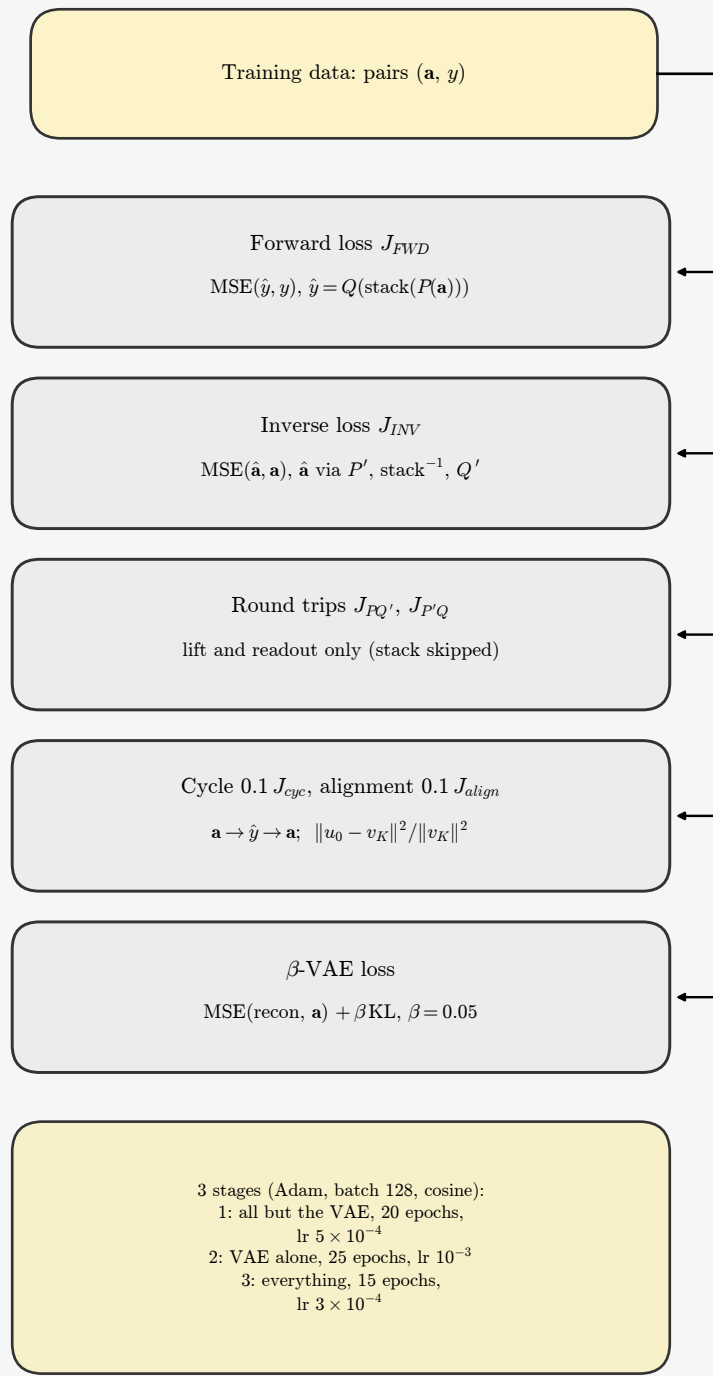
Only the coupling stack is an exact bijection; lifts and readouts are learned maps tied together by the losses.

Several designs per target; one is picked with the model's own forward direction (best predicted spectrum).

Lift P and gate L of each operator

iFNO: Fourier P: sorted encoder, query, Linear $[c, q(f), f] \rightarrow 48$ L: Fourier layer: spectral conv (48 modes) + Conv1d, GELU 188,405 parameters	iDCO: Deep Cat P: branch encoder, trunk MLP(f), query $\rightarrow$ Linear L: pointwise residual MLP block + local refinement 81,909 parameters	iGNO: Graph P: graph nodes + message passing, kernel and attention per f L: pointwise MLP + local refinement 62,210 parameters	iDNO: Deep Neural Op. P: encoder, frequency encoder, resonance query $\rightarrow$ Linear L: FiLM residual block (FiLM from a frequency embedding) + refinement 84,389 parameters
iWNO: Wavelet P: encoder, resonance query, $f \rightarrow$ Linear L: multi-level Haar wavelet block 115,213 parameters	iLNO: Laplace P: encoder $\rightarrow$ design-dependent poles, pole features, query L: learned design-independent poles, pointwise transform, refinement 83,229 parameters	iSIREN: SIREN P: $[f, q(f)] \rightarrow$ sine layers modulated by the context $\rightarrow$ Linear L: sine layer + local refinement 90,645 parameters	iSTO: Set Transformer P: resonator tokens, detuning-biased cross-attention $\rightarrow$ Linear L: frequency mixer (Conv1d, k 5, plus dilated Conv1d) 71,089 parameters

Loss functions



Recommended configuration (2 resonators): bounded design coordinates, bounded gate, binned readout, padded FFT (iFNO), cycle and alignment terms, VAE on the stage-1 estimates; f_t -sorted encoder and plate-mode features where the model has an encoder.

Parameter counts: each operator built with these options. Data: 2 resonators, 18 plate modes.